

The purpose of this project is to provide students with the opportunity to use machine learning algorithms to assist the financial industry in solving some of its strategic business problems. The project will focus on one of these two financial problems. After discussing with the instructor, students can investigate other financial problems (e.g., derivative pricing, etc.). In this case, students will be responsible for finding the dataset.

Project Description

Fraudulent Transactions

Fraud represents a major challenge within the financial industry, posing significant financial risks. It is estimated that businesses lose about 5% of their annual revenue to fraudulent activities. When applied to the global Gross World Product (GWP) of \$79.6 trillion in 2017, this equates to a potential loss of up to \$4 trillion worldwide. Detecting fraud is an ideal task for machine learning, as these algorithms can process vast amounts of transactional data, flagging any irregularities and pinpointing transactions that may be fraudulent. Furthermore, machine learning models are typically more efficient than traditional rule-based methods. By aggregating data from multiple sources and associating it with key triggers, machine learning systems can assess the likelihood of fraud or default for each transaction or customer, delivering valuable alerts and insights to financial institutions.

Loan Default

Lending plays a crucial role in the financial sector, where lenders extend loans to borrowers in exchange for repayment with interest. The lender profits only if the borrower fulfills the repayment terms. As a result, the two most pressing questions in lending are:

- What is the level of risk associated with the borrower?
- Based on this risk, is lending to the borrower a viable option?

Predicting defaults is an ideal task for machine learning, as algorithms can be trained using vast datasets of consumer information. These algorithms are capable of automating processes like matching data, spotting anomalies, and assessing whether a borrower meets the criteria for a loan. By identifying underlying patterns, these algorithms can continually monitor and analyze trends that may impact future lending and underwriting decisions.

Data¹

IEEE-CIS Fraud Detection Dataset

¹ The description of the dataset is adapted from ChatGPT-generated content.

The IEEE-CIS Fraud Detection Dataset is a publicly available dataset designed to help researchers, data scientists, and machine learning practitioners develop and test models for detecting fraudulent activities in financial transactions. This task is critical for financial institutions to prevent financial losses and protect customers from fraudulent activities. The dataset is provided by the IEEE Computational Intelligence Society (CIS) and is often used in fraud detection competitions, particularly on platforms like Kaggle.

The dataset contains anonymized transaction data. It is quite large, with millions of records and numerous features. It includes both labeled and unlabeled examples of transactions to identify fraudulent behavior among legitimate transactions. The data includes features such as:

- Transaction details (e.g., type of transaction, method of payment, transaction amount)
- User demographics (e.g., user ID, geographical location, device type)
- Device and session information (e.g., IP address, device fingerprint)
- Behavioral features (e.g., frequency of transactions)

The dataset is particularly useful for identifying fraudulent transactions based on past behavior and transaction patterns. It simulates real-world conditions where fraudulent activity needs to be detected in large-scale transactional systems. Fraudulent transactions are typically much rarer than non-fraudulent ones, which means the dataset is imbalanced. This presents a challenge for machine learning models, as they must be able to detect rare events (fraud) without being biased toward the majority class (non-fraud).

The IEEE-CIS Fraud Detection Dataset has been the focus of multiple Kaggle competitions. These competitions encourage participants to create the most effective predictive models using the data. It has led to advancements in fraud detection techniques, particularly in applying machine learning models for large-scale financial data.

It is widely used to benchmark new algorithms and techniques in machine learning, especially in the context of imbalanced data and anomaly detection. It allows for the testing and comparison of various machine learning algorithms for fraud detection, from traditional classifiers to more advanced deep learning approaches.

The dataset is publicly available on platforms such as Kaggle, where users can download the data for analysis and participate in competitions. It is typically offered in both raw and pre-processed formats to accommodate different levels of expertise in machine learning.

LendingClub Dataset

The LendingClub Dataset is a collection of data provided by LendingClub, an online peer-to-peer (P2P) lending platform. This dataset contains detailed information about loans issued on the platform, including borrower details, loan characteristics, repayment status, and other relevant financial data. It is widely used for research in finance, machine learning, credit scoring, and risk analysis.

LendingClub connects borrowers with investors by allowing individuals to borrow money directly from other individuals or institutions. The platform assesses the borrower's creditworthiness and assigns a risk profile to each loan, which is then offered to investors.

The LendingClub dataset includes data related to personal loans issued through the platform. Key fields in the dataset typically include:

- Loan information (e.g., loan amount, interest rate, term (e.g., 36 months, 60 months), loan grade, and purpose of the loan (e.g., debt consolidation, home improvement)).
- Borrower information (e.g., demographics (e.g., income, employment length, home ownership), location (e.g., state), credit scores, and debt-to-income ratio).
- Repayment status (e.g., whether the loan was paid off, whether the borrower defaulted, and whether there were any delinquencies).
- Payment history (e.g., monthly payments, the actual amount paid, and the due amount).
- Loan performance (e.g., the current status of the loan (paid, late, charged off, etc.)).

Each loan in the LendingClub dataset is typically assigned a loan grade (e.g., A, B, C) that indicates the risk level of the loan. A higher-grade loan (like A) is considered less risky, while a lower-grade loan (like D or E) suggests higher risk. This grade is used to set the interest rate for the loan and is based on factors such as the borrower's credit score, loan amount, and debt-to-income ratio.

Furthermore, each loan has a status describing whether it is:

- Fully Paid: The borrower has repaid the loan completely.
- Charged Off: The loan is considered a loss by the lender due to the borrower defaulting.
- Current: The borrower is up-to-date with their payments.
- Late: The borrower is late on payments (e.g., 30, 60, or 90 days late).

The LendingClub dataset is large, often containing hundreds of thousands of records. This makes it suitable for training machine learning models on a broad scale, especially for tasks like credit scoring, default prediction, and loan performance analysis. The dataset may be imbalanced, especially when dealing with loan defaults. This can create challenges for machine learning models (e.g., in terms of bias and overfitting).

The dataset is often made available through various sources, such as:

- LendingClub's website (for historical data).
- Kaggle (where the dataset is used for competitions and learning).
- Public repositories on GitHub or other data science platforms.

Instructions

1. General analysis
 - a. Provide a general overview of the business problem.
 - b. Describe the challenges.
 - c. Explain measures employed by financial institutions to address the problem.
2. Use single or multiple datasets. Search or use the datasets from the following:
 - a. Use datasets attached (recommended option) OR
 - b. OR Search for datasets on Kaggle.com
 - c. OR Find datasets from other sources.
3. Exploratory data analysis
 - a. Descriptive statistics
4. Data preparation
 - a. Data cleaning
 - b. Data wrangling
 - c. Feature engineering
 - d. Data visualization
5. Evaluate ML models
 - a. Use at least 8 models (e.g., logistic regression, clustering, dimension reduction, decision tree, random forest, KNN, XGboost).
 - b. Evaluation metrics.
 - i. Accuracy score
 - ii. Mean Squared errors.
 - iii. R-squared
 - iv. Others
 - c. Compare models.
 - d. Select the best model.
6. Model tuning and grid search
7. Finalize the model
 - a. Results on test data
 - b. Variable intuition/feature selection
8. Provide a detailed analysis or explanation of how the model results help to solve the business problem.
 - a. Explain the results.
 - i. How good is the model?
 1. Explain the accuracy score or mean-square-error.
 2. Is the score good or not, why, and why not?
 3. How can you improve the model's performance?
 - ii. Explain how the model you created helps solve the business problem(s)
 - iii. What would you do differently if you had to build/deploy this algorithm for a prospective employer?
9. Document some of the weaknesses of the model.
10. Provide suggestions on how to improve the model and the results.

Project Deliverables

Milestone 1 (2 to 5 pages Word Document, due on 02/01/2026): Review of literature

- Key takeaways from the reading
 - Research objectives
 - Data preparation
 - Feature engineering
 - Model performance metrics
 - Models executed
 - Models comparison

Milestone 2 (Jupyter Notebook, due 03/01/2026): Project proposal

- General analysis of the business problem
- Status of the data collection
- Exploratory data analysis
- Data preparation

Milestone 3 (Jupyter Notebook, due 03/29/2026): Preliminary results (half of the models)

- Evaluate ML models (at least 4 models)

Milestone 4 (Jupyter Notebook + Streamlit Web App Link, due 04/19/2026): Final report

- Evaluate ML models (at least 8 models)
- Model tuning and grid search
- Finalize the model
- Provide a detailed analysis or explanation of how the model results help to solve the business problem.
- Document some of the weaknesses of the model.
- Provide suggestions on how to improve the model and the results.